Richard Zhang

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Education

Carnegie Mellon University, School of Computer Science — Pittsburgh, PA
B.S. in Computer Science (Machine Learning Concentration) — *Expected Graduation: May 2027*

Experience

Amazon Web Services (AWS) — May 2026 – Aug. 2026
Software Development Engineer Intern — *Seattle, WA*

- Built an agentic system on Amazon Bedrock AgentCore that conversationally completes enterprise SSO onboarding for QuickSight across four identity providers (Okta, PingFederate/PingOne, Microsoft Entra ID, Google Workspace), replacing manual console configuration which reduced enterprise SSO onboarding time by 50%
- Built mobile and web Build Verification Tests into the release pipeline that sign in as a real enterprise identity and exercise chat, auth, and artifact flows on physical devices to automatically catch regressions
- Shifted regression discovery for auth and identity flows from post-release, user-reported issues to pre-release, automated detection
- Tools Used: Amazon Bedrock AgentCore, AWS CDK, AWS Device Farm, Python, TypeScript, GitFarm, Linux

Every Prosperity Properties — Aug. 2025 – Present
AI Software Engineer — *Fairfax, VA*

- Built an AI-powered home search app enabling listing inquiries, showing scheduling, and price analysis with instant client responses
- Deployed and managed backend services on Microsoft Azure using App Service and Azure Database for PostgreSQL to support scalable production workloads
- Implemented OAuth2 authentication and built a custom web scraper to preprocess and store structured property data in PostgreSQL
- Tools Used: Python, OpenAI, PostgreSQL, Microsoft Azure, FastAPI, OAuth2, Playwright

Lightspeed LA — May 2025 – Jul. 2025
Software Engineer — *Irvine, CA*

- Trained a reinforcement learning agent using Proximal Policy Optimization (PPO) to enable a humanoid model to perform stable bipedal walking and recovery from perturbations for use in next-generation video games
- Transitioned simulation environments from Isaac Gym to Genesis to improve stability and training scalability
- Tools Used: Genesis Embodied AI, PyTorch, CUDA, NumPy, Python, Tensorboard, Git, Linux

Projects

Presenter.AI - Web Application (Hackathon)

- Built a real-time web app providing AI-powered live feedback on speech clarity, pacing, and filler words using React, Node.js/Express, Socket.IO, and the OpenAI API
- Tools Used: React, Node.js, Express, Socket.IO, Git, Linux, WebSockets, Speech APIs

Activities

Carnegie Mellon Racing Driverless — Aug. 2024 – Jun. 2026
Path Planning Captain — *Pittsburgh, PA*

- Leading autonomous navigation stack development (mapping, localization, motion planning) for a 110-member team competing in Formula SAE Michigan
- Developed AI-driven raceline generation algorithms using SVMs to maximize path smoothness and minimize curvature for optimal lap performance
- Tools Used: C++, Python, ROS2, scikit-learn, OpenCV, Eigen, NumPy, Git, Linux

Skills

Skills: Python, C++, Java, JavaScript, React, Node.js, Express, PostgreSQL, HTML/CSS, Git, Linux, Microsoft Azure, AWS, Amazon Bedrock, REST APIs, WebSockets, CI/CD, Agile Development